

Linear equations with numerical fractions

While TSM 099 studies Topic 29. Review: Topic 28.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: combine fractions on each side

1. Solve and check in the original equation.

$$\frac{6x - 5}{6} = \frac{2x - 1}{4}$$

$$4(6x - 5) = 6(2x - 1)$$

$$x = \frac{7}{6}$$

Check by substituting into the original equation.

2. Solve and check in the original equation.

$$\frac{-8x + 3}{20} = \frac{x + 6}{8}$$

$$8(-8x + 3) = 20(x + 6)$$

$$x = -\frac{8}{7}$$

Check by substituting into the original equation.

3. Solve and check in the original equation.

$$\frac{6x + 5}{6} = \frac{x + 1}{2}$$

$$2(6x + 5) = 6(x + 1)$$

$$x = -\frac{2}{3}$$

Check by substituting into the original equation.

4. Solve and check in the original equation.

$$\frac{-13x - 16}{8} = \frac{-x - 3}{2}$$

$$2(-13x - 16) = 8(-x - 3)$$

$$x = -\frac{4}{9}$$

Check by substituting into the original equation.

5. Solve and check in the original equation.

$$\frac{x - 1}{6} = \frac{-2x - 6}{3}$$

$$3(x - 1) = 6(-2x - 6)$$

$$x = -\frac{11}{5}$$

Check by substituting into the original equation.

6. Solve and check in the original equation.

$$\frac{-2x + 4}{3} = \frac{-x - 1}{5}$$

$$5(-2x + 4) = 3(-x - 1)$$

$$x = \frac{23}{7}$$

Check by substituting into the original equation.

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Core: numerical denominators on both sides

7. Solve and check in the original equation.

$$\frac{9x - 5}{6} = \frac{x + 6}{6}$$

$$6(9x - 5) = 6(x + 6)$$

$$x = \frac{11}{8}$$

Check by substituting into the original equation.

8. Solve and check in the original equation.

$$\frac{x + 4}{2} = \frac{x + 4}{5}$$

$$5(x + 4) = 2(x + 4)$$

$$x = -4$$

Check by substituting into the original equation.

9. Solve and check in the original equation.

$$\frac{12x - 3}{10} = 0$$

$$1(12x - 3) = 10(0)$$

$$x = \frac{1}{4}$$

Check by substituting into the original equation.

10. Solve and check in the original equation.

$$\frac{-2x + 1}{8} = \frac{-x + 2}{2}$$

$$2(-2x + 1) = 8(-x + 2)$$

$$x = \frac{7}{2}$$

Check by substituting into the original equation.

11. Solve and check in the original equation.

$$\frac{-2x - 5}{3} = -1$$

$$1(-2x - 5) = 3(-1)$$

$$x = -1$$

Check by substituting into the original equation.

12. Solve and check in the original equation.

$$\frac{-2x + 1}{3} = \frac{1}{4}$$

$$4(-2x + 1) = 3(1)$$

$$x = \frac{1}{8}$$

Check by substituting into the original equation.

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Core: signed numerators and fractions

13. Solve and check in the original equation.

$$\frac{3x - 11}{4} = 1$$

$$1(3x - 11) = 4(1)$$

$$x = 5$$

Check by substituting into the original equation.

14. Solve and check in the original equation.

$$\frac{-11x - 11}{40} = 1$$

$$1(-11x - 11) = 40(1)$$

$$x = -\frac{51}{11}$$

Check by substituting into the original equation.

15. Solve and check in the original equation.

$$\frac{x + 7}{12} = \frac{-x + 5}{3}$$

$$3(x + 7) = 12(-x + 5)$$

$$x = \frac{13}{5}$$

Check by substituting into the original equation.

16. Solve and check in the original equation.

$$\frac{-19x + 16}{24} = \frac{-x + 2}{3}$$

$$3(-19x + 16) = 24(-x + 2)$$

$$x = 0$$

Check by substituting into the original equation.

17. Solve and check in the original equation.

$$\frac{-7x - 20}{12} = \frac{-2x + 3}{2}$$

$$2(-7x - 20) = 12(-2x + 3)$$

$$x = \frac{38}{5}$$

Check by substituting into the original equation.

18. Solve and check in the original equation.

$$\frac{10x + 7}{12} = \frac{2x + 1}{6}$$

$$6(10x + 7) = 12(2x + 1)$$

$$x = -\frac{5}{6}$$

Check by substituting into the original equation.

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Extension: mixed fraction equations

19. Solve and check in the original equation.

$$\frac{-2x + 1}{3} = \frac{x - 4}{4}$$

$$4(-2x + 1) = 3(x - 4)$$

$$x = \frac{16}{11}$$

Check by substituting into the original equation.

20. Solve and check in the original equation.

$$\frac{-5x - 6}{12} = \frac{5}{4}$$

$$4(-5x - 6) = 12(5)$$

$$x = -\frac{21}{5}$$

Check by substituting into the original equation.

21. Solve and check in the original equation.

$$\frac{6x + 17}{12} = \frac{4}{3}$$

$$3(6x + 17) = 12(4)$$

$$x = -\frac{1}{6}$$

Check by substituting into the original equation.

22. Solve and check in the original equation.

$$\frac{x + 3}{5} = \frac{2x + 5}{4}$$

$$4(x + 3) = 5(2x + 5)$$

$$x = -\frac{13}{6}$$

Check by substituting into the original equation.

23. Solve and check in the original equation.

$$\frac{-x - 3}{4} = \frac{-2x + 4}{3}$$

$$3(-x - 3) = 4(-2x + 4)$$

$$x = 5$$

Check by substituting into the original equation.

24. Solve and check in the original equation.

$$\frac{-5x + 6}{6} = -x - 1$$

$$1(-5x + 6) = 6(-x - 1)$$

$$x = -12$$

Check by substituting into the original equation.